

Jordy Van Landeghem

Applied Research Scientist · Reliable Multimodal Agentic Systems

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Portfolio · LinkedIn · GitHub · Google Scholar

Profile

AI/ML research scientist and engineer with 10+ years building production systems and a Ph.D. from KU Leuven. Experience spans multimodal architectures, LLM/VLM post-training, distributed training, agentic AI, and evaluation, taking research from experiments into production. At Instabase, owned InstaLLM's supervised fine-tuning and post-training framework for enterprise model adaptation. Creator of DUDE (ICCV 2023); co-author of MADQA (ICML 2026 Oral).

Experience

Probably Approximately Human BV

Founder & AI Consultant

Remote [Belgium]

2026–present

- Advise enterprises on AI adoption and innovation; deliver custom models and systems tailored to their specific needs.
- Work as a **forward-deployed applied researcher** in intelligent document processing (IDP), combining enterprise delivery with benchmark leadership (DUDE) and research on reliable multimodal agents. **DAS 2026 keynote speaker**: “Parse, Reflect, Retrieve, Compile: An Agent Stack for Enterprise Document AI.”

Instabase

Senior Software Engineer, Machine Learning

Remote

2024–September 2026

- Built distributed LLM training and fine-tuning pipelines for **multi-GPU workloads** using PyTorch FSDP and DeepSpeed.
- Owned **InstaLLM**, an on-premises LLM solution, and built its supervised fine-tuning and post-training framework for enterprise model adaptation.
- Designed and shipped **DocumentReActAgent** — agentic self-correction loop scaled to 1M+ documents per day; lifted accuracy 94%→100% and straight-through processing 30%→100% on a flagship enterprise dataset, and 85%→92%+ on another, resolving the hard edge cases where document automation typically hits its ceiling.
- Built **evaluation & benchmarking framework** (multi-provider, JSON schema abstraction, LLM-as-a-Judge, principled confidence estimation methods); used for new model release evaluation and production decision-making.
- Authored **PRD for Unified Extractor v2** architecture: schema / state / prompts / engines / orchestration separation; protocol-based plug-and-play intelligence — adopted as the core backend architecture of Instabase's general document extraction product.
- Technical lead of small, high-agency **Agent Mode** “tiger” team (<10); mentored engineers across Project Accuracy, Agent Mode, and AXIS teams.
- Delivered successful prototypes in **in-house company hackathons**: multimodal retrieval & answer grounding (IBHack I); agentic document parsing “Thinking with Documents” (IBHack II) — both prototypes became integral parts of the product roadmap.

Contract.fit (A Netcall Company)

Artificial Intelligence / NLP Engineer

Brussels, Belgium

June 2017–March 2024

- Led a small, high-agency team (<10) on end-to-end **Document AI engineering** (NLP + CV) for insurance, finance, and legal domains across a SaaS product used by enterprise clients.
- Designed and shipped **production-grade ML pipelines** for document classification and information extraction; maintained and evolved them over 7 years.
- Secured **4 Flemish innovation grants** (VLAIO, total multi-year R&D funding) as lead researcher; co-wrote all grant applications.
- Supervised **11 Master's AI/CS thesis internships** (KU Leuven, VUB); mentored junior engineers.
- Drove adoption of academic advances (DUDE benchmark, uncertainty estimation) into scalable product features.

Nuance Communications

Research Intern — Language Modelling

Aachen, Germany

2017

- Researched advanced regularisation techniques for RNN language models; implemented biLSTM character-based word embeddings.

Oracle

Research Intern — NLP for Dialogue Systems

Barcelona, Spain

2016–2017

- Investigated Seq2Seq neural networks for chatbot and virtual assistant technology.

Education

KU Leuven

Ph.D. in Engineering Sciences (Computer Science)

Thesis: *Intelligent Automation for AI-driven Document Understanding*

Supervisors: Prof. M.-F. Moens, Prof. M.B. Blaschko TA: Data Structures & Algorithms (2022–23)

Belgium

2020–2024

KU Leuven

M.Sc. Artificial Intelligence: Speech & Language Technology (Magna Cum Laude)

Thesis: *Sequence-to-Sequence Learning for End-to-End Dialogue Systems*

Belgium

2015–2017

KU Leuven

M.A. Linguistics: Corpus & Usage-based Methodologies (Magna Cum Laude)

Belgium

2014–2015

Selected Publications

2026: Borchmann, L., **Van Landeghem, J.** et al. *Strategic Navigation or Stochastic Search? How Agents and Humans Reason Over Document Collections (MADQA)*. ICML 2026. *Oral (top 0.5%)*.

2025: Zhou, H., **Van Landeghem, J.** et al. *A Novel Characterization of the Population AURC and Rates of Finite Sample Estimators*. ICML 2025.

2025: Biescas, A., Biswas, S., Lladós, J., **Van Landeghem, J.** *Where Layout Meets Language*. ICDAR 2025.

2024: **Van Landeghem, J.** et al. *Beyond Document Page Classification: Design, Datasets, and Challenges*. WACV 2024.

2024: **Van Landeghem, J.** et al. *DistilDoc: Knowledge Distillation for Visually-Rich Document Applications*. ICDAR 2024.

2023: **Van Landeghem, J.** et al. *Document Understanding Dataset and Evaluation (DUDE)*. ICCV 2023. *154 citations*.

2022: **Van Landeghem, J.**, Blaschko, M., Anckaert, B., Moens, M.-F. *Benchmarking Scalable Predictive Uncertainty in Text Classification*. IEEE Access 2022.

2020: **Van Landeghem, J.** et al. *Predictive Uncertainty for Probabilistic Novelty Detection in Text Classification*. ICML Workshop on Uncertainty & Robustness in Deep Learning, 2020.

Grants & Funding

2026: BELSPO R&D innovation grant — *pending*: Multimodal AI-Agent Architectures for Automated Document Intelligence

2019: Personal Baekeland PhD Fellowship, VLAIO (HBC 2019.2604)

2017–2021: 3× Company Innovation Grants, VLAIO (HBC.2017.0264, HBC.2019.2376, HBC.2021.0787)

Selected Talks

DAS 2026: Keynote: *Parse, Reflect, Retrieve, Compile: An Agent Stack for Enterprise Document AI*

WACV 2024: Oral presentation: *Beyond Document Page Classification*

Adobe Research: Invited seminar (Oct 2023): *DUDE — what's next?*

CVC/UAB: Invited seminar (Jun 2023): Calibration primer

ICDAR 2023: Oral: DUDE competition

Technical Skills

AI/ML: LLM/VLM adaptation, distributed training, SFT, DPO, agentic systems, RAG, document AI, evaluation and benchmarking, uncertainty estimation, knowledge distillation

Frameworks: PyTorch, FSDP, DeepSpeed, Hugging Face Transformers, LangChain, LangGraph, LlamaIndex, scikit-learn

Production: FastAPI, Pydantic, Docker, Kubernetes, MLflow, AWS, GCP, Azure, CI/CD; FAISS, Pinecone, Weaviate

Models & APIs: Llama, Mistral, Qwen, Gemma; OpenAI, Anthropic, Google Gemini

Programming: Python, JavaScript, Bash, SQL, LaTeX

Languages

Dutch (native), English (C2), Spanish (C1), French (C1), German (B2), Portuguese (A2)